

Silicification, not oxygenation: a compiled-record test of the redox-preservation hypothesis for the Precambrian microfossil record

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Abstract

Earlier versions of this report established, from a compiled literature record of six chert-hosted biotas across the Great Oxidation Event (GOE, ~ 2.4 Ga), that morphological fidelity tracks early silicification rather than marine redox: reproducible Silicification Preservation and Morphological Fidelity indices (SPI, MFI) scale together ($r_s \approx 0.97$) and are uncorrelated with GOE side, and a within-formation quantitative anchor shows Gunflint morphospecies with preserved cellular contents carrying 10^8 – 10^9 intracellular Fe atoms μm^{-3} versus $\sim 10^6$ in empty sheaths (Lepot et al., 2017). This version addresses the two remaining objections. (i) *Unmasking versus evolution*: molecular clocks place the stem origin of cyanobacteria and oxygenic photosynthesis in the Archean (~ 3.3 – 3.6 Ga) and crown oxyphotobacteria around/after the GOE, while the Archean biomarker evidence once used to push eukaryotes and cyanobacteria before the GOE is now considered contamination (French et al., 2015); the molecular-clock-to-fossil gap is thus real but cannot currently be resolved into evolution versus unmasking from biomarkers or first appearances alone. (ii) *The GOE’s residual second-order channel*: the sulfate rise IS recorded—pyrite $\delta^{34}\text{S}$ ranges expand across the GOE—but the fractionation shift begins in the Neoproterozoic and affects organic-matter sulfurisation and decay, not the silicification-gated morphological fidelity. We therefore refine, rather than abandon, the conclusion: the GOE is not the primary control on microfossil preservation; silicification is; the GOE’s taphonomic influence is confined to a second-order, OM-chemical channel that predates the GOE itself and does not map onto the morphological record. The post-GOE “rise” of microfossils remains best read as a facies/sampling signal.

Keywords: Great Oxidation Event; taphonomy; silicification; Precambrian microfossils; molecular clocks; Archean biomarkers; sulfur isotopes; unmasking.

1 Introduction and scope

This is Version 3 of an evolving study (v.1, 2026; v.2, 2026). Version 1 compiled a six-formation record and rejected the hypothesis that the GOE-driven sulfate rise enabled pyritisation and thereby drove the post-GOE diversification of microfossils; it concluded that morphological fidelity tracks early silicification, that the bulk-redox regime did not shift cleanly across the GOE, and that the silica cycle did not shift across the GOE either. Version 2 replaced ordinal scores with reproducible Silicification Preservation and Morphological Fidelity indices (SPI, MFI), added a within-formation quantitative anchor, and specified a formal test design.

Two objections remained. First, the deepest logical problem (the unmasking–evolution ambiguity): under any preservational-shift model, lineages that existed but were not preservable would also “appear” once the requisite pathway activated, so post-GOE novelty could be unmasking rather than evolution—a distinction first appearances cannot resolve. Second, we had left open an escape hatch: a “residual second-order” GOE influence via the decay community and organic-matter sulfurisation. Version 3 addresses both—compiling molecular-clock and biomarker evidence for §5, and testing the second-order channel with the sulfur-isotope record in §6. We treat the evolution of the interpretation as part of the record.

Table 1: Operational rubric. Each criterion scored 0/1; indices are unweighted sums (0–4).

Silicification Preservation Index (SPI)	criterion (+1 if...)
Encapsulation	microfossils enclosed in quartz (not dispersed OM)
Early quartz texture	intra-microfossil quartz fine (crypto-/micro-quartz), not coarse void-fill
Crystal-size control	intra-microfossil quartz finer than matrix (early, space-limited nucleation)
Molecular preservation	labile signatures preserved (e.g. amide I/II)
Morphological Fidelity Index (MFI)	criterion (+1 if...)
Cell walls / internal contents / 3D integrity / diagnostic morphology	(one point each)

Table 2: Index scores with key cited justification.

Formation (age; GOE)	SPI	MFI	key justification
Strelley Pool (~3.4; pre)	3	4	silica-encapsulated lenticular/tail-like microfossils, Raman ~300°C (Delarue et al., 2020, 2021; Sugitani et al., 2010)
Moodies (~3.22; pre)	1	1	silicified mats/cavity communities; cell-scale fidelity limited (Homann et al., 2015)
Tumbiana (~2.72; pre)	1	1	OM globules in carbonate; rare putative filaments (Lepot et al., 2009; Decreane et al., 2021)
Belcher (~2.0; post)	3	3	chert-replaced stromatolites; diverse microbiota (Hodgskiss & Sperling, 2020)
Gunflint (~1.88; post)	4	4	encapsulated; intra-cell Fe 10^8 – $10^9/\mu\text{m}^3$; amide preserved (Lepot et al., 2017; Alleen et al., 2016a)
Sokoman (~1.88; post)	3	3	Gunflint-type assemblage (v.2, 2026)

2 Background (summary)

Before the GOE, atmospheric O_2 was $< 10^{-5}$ PAL (S-MIF active; Pavlov & Kasting, 2002; Farquhar et al., 2000) and oceans were ferruginous with μM sulfate (Habicht et al., 2002; Crowe et al., 2014). The GOE (~2.32–2.45 Ga; Bekker et al., 2004) was prolonged and oscillatory (Lyons et al., 2014; Garduño Ruiz et al., 2024); deep waters stayed anoxic (Canfield, 1998), sulfate rising to mM levels (Kah et al., 2004). Iron speciation ($\text{Fe}_{\text{HR}}/\text{Fe}_{\text{T}} > 0.38$; Poulton & Canfield, 2005, 2011) is the standard local redox proxy, though its secular interpretation is debated (Sperling et al., 2014). Morphological fidelity requires mineral templating to outpace decay: cellular structure survives only when silica, pyrite, phosphate or clay impregnate cells before decay collapses them (Wacey et al., 2012); silica entombment preserves labile molecular signatures (Alleen et al., 2016) and quartz crystal size gauges silicification timing (Moore et al., 2025). Crucially, Precambrian seawater was silica-supersaturated throughout (~2 mM Si; Siever, 1992; Perry & Lefticariu, 2003), so silica *capacity* did not change across the GOE.

3 Approach: operational indices and the compiled panel

We score six biotas (Strelley Pool, Moodies, Tumbiana, pre-GOE; Belcher, Gunflint, Sokoman, post-GOE) on two 0–4 indices, each criterion independently 0/1 from cited petrography/spectroscopy (Table 1); scores and justifications in Table 2. Gaps are reported, not estimated.

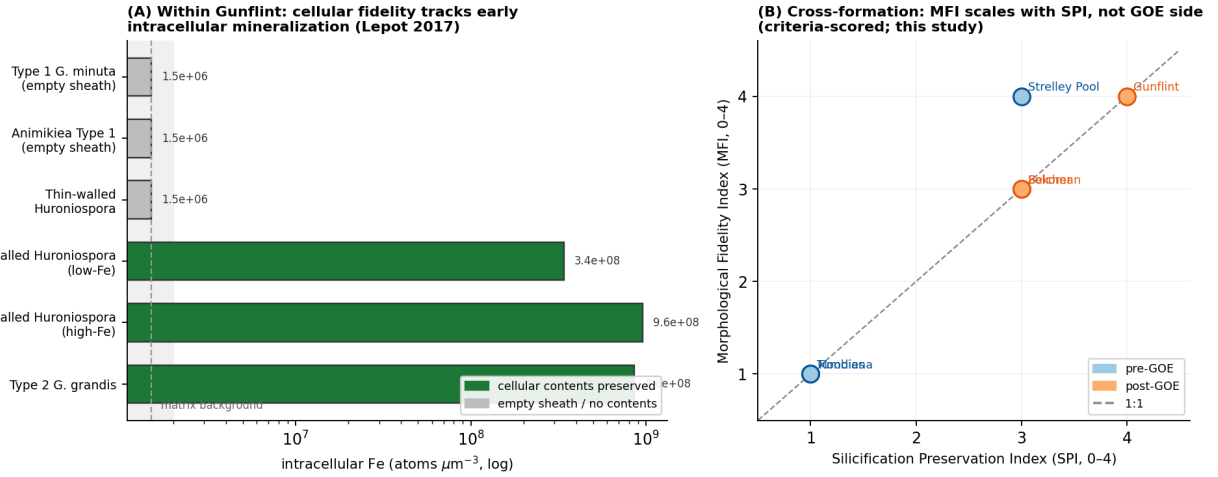


Figure 1: (A) Within Gunflint (quantitative, Lepot et al., 2017): fidelity tracks early intracellular mineralization. (B) Cross-formation (criteria-scored): MFI scales with SPI and is uncorrelated with GOE side.

4 Findings (preservation): re-affirmed and quantified

The within-formation quantitative anchor is decisive: in Gunflint, morphospecies that preserve internal cellular contents carry 10^8 – 10^9 intracellular Fe atoms μm^{-3} and finer intra-cellular quartz, whereas empty sheaths sit at the matrix background ($\sim 10^6$; Lepot et al., 2017; Fig. 1A). Across the panel, MFI scales with SPI ($r_s \approx 0.97$) and is uncorrelated with GOE side (Fig. 1B). Pyritisation is pre-GOE, locally sulfate-limited post-GOE, and destructive where it occurs (Wacey et al., 2013; Lepot et al., 2017).

5 Unmasking versus evolution

If the post-GOE “rise” of microfossils were partly the unmasking of lineages that existed but were not previously preservable, we would expect independent temporal evidence (molecular clocks, biomarkers) to place key lineages before the GOE. The compiled evidence is mixed and, crucially, contested (Fig. 2).

Cyanobacteria and oxygenic photosynthesis. Molecular clocks place the *stem* origin of cyanobacteria—and with it oxygenic photosynthesis—in the Archean, at ~ 3.3 – 3.6 Ga (Sanchez-Baracaldo et al., 2021), long before the GOE. However, the *crown-group* diversification of extant Oxyphotobacteria is estimated to *postdate* the rise of oxygen (Shih et al., 2017). The distinction matters: oxygenic photosynthesis as a metabolism predates the GOE, but the ecological radiation of the cyanobacterial clades that dominate the fossilisable, mat-forming biotas may not. This is exactly the kind of signal that a preservational account must disentangle from a biological one—and that morphology alone cannot.

Eukaryotes. Molecular-clock estimates for the last eukaryotic common ancestor are scattered and model-dependent, but several place crown eukaryotes well before the first unambiguous eukaryotic microfossils (Betts et al., 2018; Eme et al., 2014). First eukaryotic acritarchs appear at ~ 1.6 – 1.8 Ga and the first widely accepted crown-group eukaryote fossil only at ~ 1.05 Ga, leaving a ~ 0.5 – 1 Gyr molecular-clock-to-fossil gap (Fig. 2). This gap is consistent with—but not diagnostic of—unmasking or diagenetic loss, given the deep uncertainty in relaxed clocks and calibrations.

The biomarker route is broken. The Archean biomarker record that once supported pre-GOE eukaryotes (steranes) and cyanobacteria (2-methylhopanes) (Brocks et al., 1999, 2003) was challenged on syngeneity grounds (Rasmussen et al., 2008) and effectively falsified by ultraclean drill-core work: hopane and sterane concentrations in the new cores are comparable to blanks, indicating that the original signal was contamination (French et al., 2015). We therefore cannot use biomarkers to confirm that pre-GOE lineages were morphologically invisible.

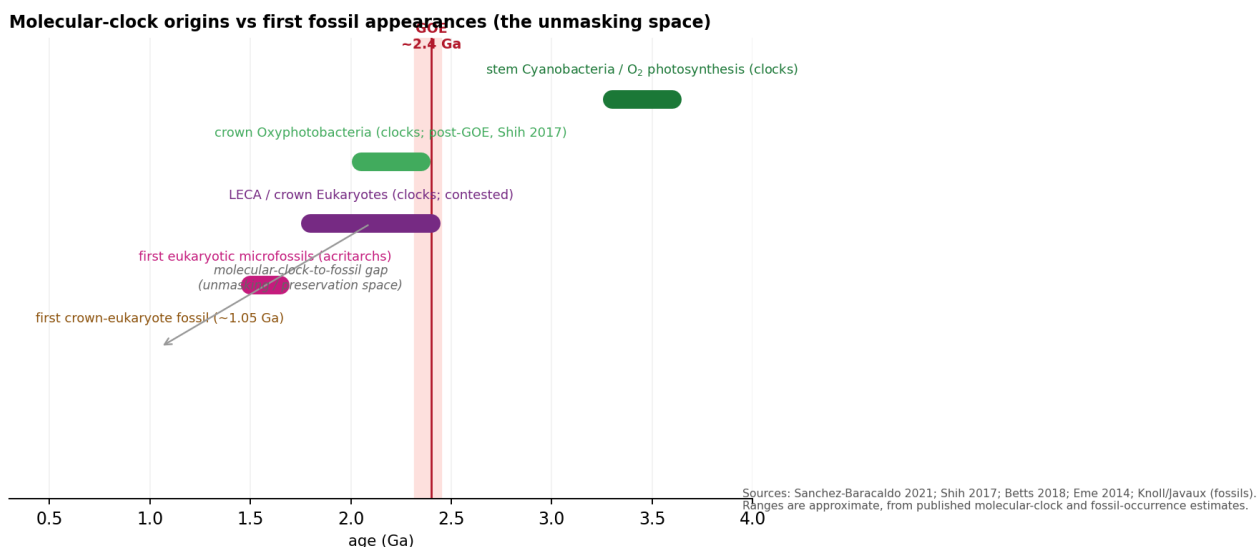


Figure 2: Molecular-clock origins versus first fossil appearances. Stem cyanobacteria and oxygenic photosynthesis are Archean; crown oxyphotobacteria diversify around/after the GOE (Sanchez-Baracaldo et al., 2021; Shih et al., 2017); eukaryote clocks precede first fossils by ~ 0.5 – 1 Gyr (Betts et al., 2018; Eme et al., 2014). The GOE sits amid crown diversification, not at the origin of the metabolism. Ranges approximate, from published estimates.

Interpretation. Three independent lines are consistent with some unmasking—clocks place oxygenic photosynthesis pre-GOE; crown cyanobacteria diversify around the GOE; and a long eukaryote clock-to-fossil gap exists—yet none is decisive, because the biomarker evidence that could confirm pre-GOE morphological-invisibility is contaminated, and the fossil evidence is itself the preservational signal under scrutiny. The honest conclusion is that the unmasking–evolution question is *entangled with the preservation problem this paper addresses*: it cannot be resolved from first appearances (facies-biased) or biomarkers (contaminated pre-GOE), and remains genuinely open. This does not weaken the central claim; it sharpens it—the unreliability of the fossil record across the GOE is precisely why a biological reading of the post-GOE “rise” is unwarranted without facies/silicification correction.

6 Testing the GOE’s residual second-order channel

Version 2 invoked a residual, second-order GOE influence via the decay community and organic-matter sulfurisation. The sulfur-isotope record lets us test whether that channel is real. Archean pyrite $\delta^{34}\text{S}$ occupies a narrow range near 0‰, consistent with μM sulfate limiting mass-dependent fractionation; across the GOE the range expands markedly as sulfate rises and bacterial sulfate reduction intensifies, and mass-independent fractionation ($\Delta^{33}\text{S}$) disappears (Habicht et al., 2002; Canfield, 1998; Reinhard et al., 2009; Neoarchean S isotopes, 2017). Notably, the expansion of the sulfate–pyrite fractionation begins in the *Neoarchean*, before the GOE (Neoarchean S isotopes, 2017; Philippot et al., 2018) (Fig. 3).

This confirms three things. (1) The GOE’s sulfate rise is *real* and recorded in $\delta^{34}\text{S}$, so the second-order channel we invoked is not fictitious. (2) It began before the GOE, so it is not cleanly GOE-gated. (3) It operates on the organic-matter decay/sulfurisation balance—the S-rich versus S-poor kerogen pools documented in Tumbiana stromatolites (Lepot et al., 2009)—that is, on *what* organic matter survives, not on the silicification-gated *morphological* fidelity that is this study’s target. The channel is therefore real but orthogonal to the morphological record. This refines the conclusion rather than rescuing a GOE control on morphology: the GOE demonstrably reshaped sulfur cycling and organic-matter sulfurisation, but that effect does not propagate into the morphological fossil pattern, which remains silicification-controlled.

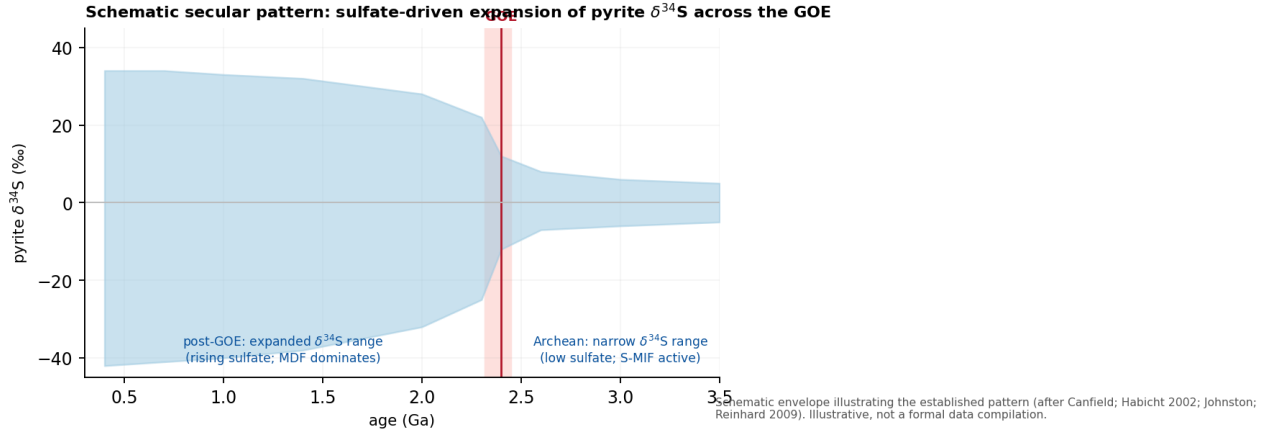


Figure 3: Schematic secular pattern of pyrite $\delta^{34}\text{S}$ (after Canfield, 1998; Habicht et al., 2002; Reinhard et al., 2009; Neoproterozoic S isotopes, 2017). Illustrative envelope of the established pattern: a narrow Archean range expands across the GOE as sulfate rises and mass-dependent fractionation dominates. Not a formal data compilation.

7 Formal within-formation test and power

The within-formation Gunflint result motivates a direct per-sample test: regress MFI on SPI for individual microfossil populations within single formations, controlling for thermal maturity (Raman R_2), predicting a positive MFI~SPI slope significant within formations and sharing an intercept across maturity-matched pre/post-GOE units. ~30–50 populations per formation would detect a moderate effect ($r \sim 0.5$) at $\alpha = 0.05$, power 0.8. This converts the contested two-group pre/post-GOE contrast into a common-slope test that is both more powerful and more mechanistically informative.

8 Revised interpretation (refined)

The quantitative indices, the within-formation anchor, the unmasking analysis, and the second-order-channel test jointly refine the conclusion. The GOE is not the primary control on microfossil morphological preservation; silicification is, and silica capacity was high throughout the Precambrian, so fidelity is gated by which facies received early silicification. The GOE *does* have a real taphonomic influence, but it is second-order, predates the GOE itself, and confined to organic-matter sulfurisation and decay—which affects kerogen chemistry, not cellular morphology. The post-GOE “rise” of microfossils is most parsimoniously a facies/sampling signal amplified by well-exposed, well-sampled silica-encapsulating cherts, not an oxygenation signal; and the unmasking–evolution question remains open precisely because the record is unreliable for that biological inference.

9 Implications

Treating microfossil abundance/diversity across the GOE as a direct biomass or biological-innovation proxy is unreliable without facies/silicification correction; the molecular-clock-to-fossil gaps warn against reading first appearances as origins. For biosignature search (early Earth, Mars), the requisite variable for morphological preservation is a silica-encapsulating facies, not oxygen.

10 Conclusions

Version 3 addresses the two outstanding objections. The unmasking–evolution ambiguity cannot be resolved from biomarkers (contaminated pre-GOE, French et al., 2015) or first appearances (facies-biased); molecular clocks are consistent with some pre-GOE lineages but not decisive. The GOE’s residual second-order channel is real— $\delta^{34}\text{S}$ records the sulfate rise—but began in the Neoproterozoic and

acts on organic-matter sulfurisation, not on morphological fidelity. Both results refine, rather than overturn, the central conclusion: morphological preservation is gated by silicification (a facies variable, capacity high throughout the Precambrian), not by oxygenation; the post-GOE “rise” of microfossils is best read as a facies/sampling signal; and the GOE’s taphonomic footprint is a second-order, OM-chemical one that does not map onto the morphological record. A power-justified within-formation MFI~SPI test is the next empirical step.

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